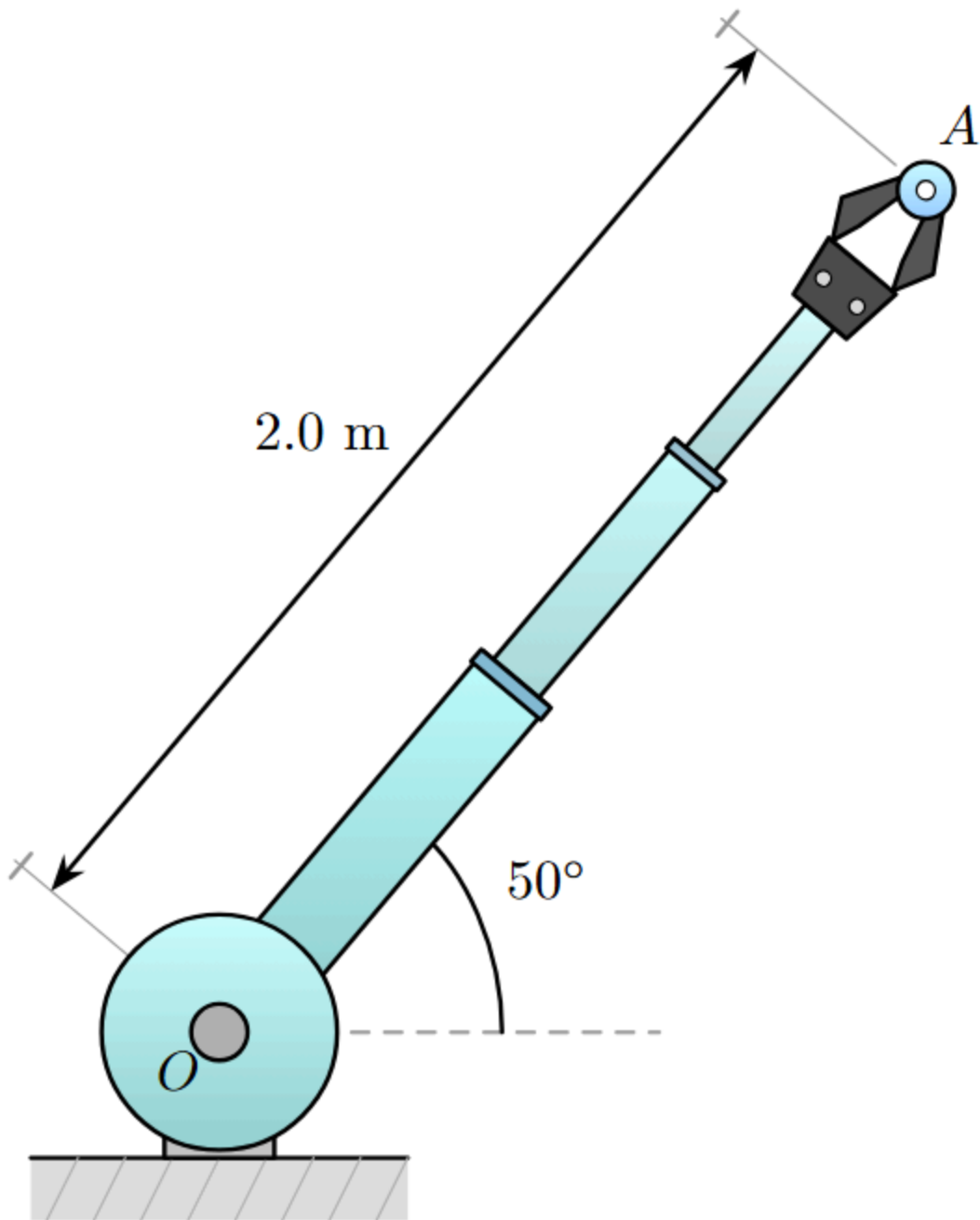


Lesson 5: 2D motion with polar coordinates

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1.

/1

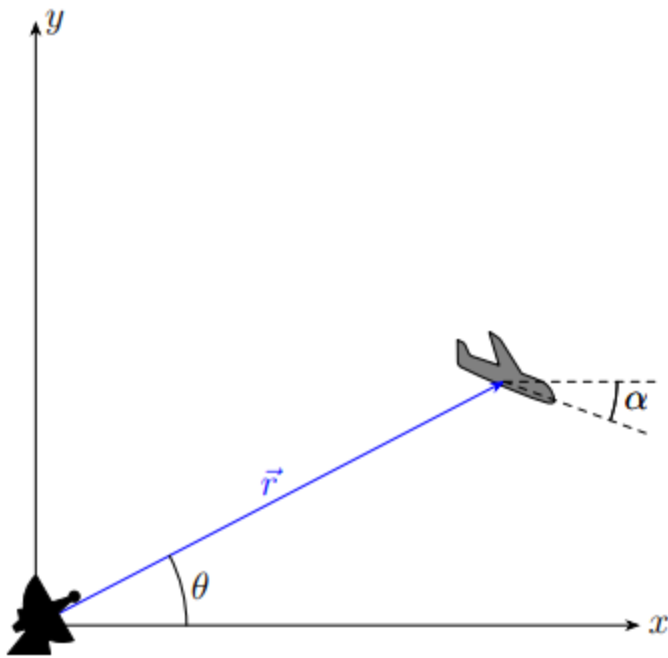


At the instant shown, the length of the robotic arm is decreasing at a constant rate of 0.2 m/s , while being lowered by a constant rate of 0.5 rad/s . Determine the magnitude of the acceleration of point A .

Answer: _____

2.

/1



An airplane is being monitored by a radar tracking station. When the plane is at the location shown, the distance and angle of elevation of the plane are, $r=12,600$ ft and $\theta=30^\circ$, respectively. Two seconds later, the plane is sighted at $r=13,650$ ft and $\theta=25^\circ$. Using this two second interval, approximate the speed and pitch angle, α , of the plane.
